Meeting 4B — Runtime State Architecture

Part 2: Recorder & Playback State Machines

Purpose

This document defines the **runtime states** governing recording and playback.

Unlike the audio engine, these are true firmware states that the workstation explicitly enters and exits.

Only one recorder state and one playback state may be active at any instant.

Runtime State Machine 3 — Recorder

The recorder subsystem manages the complete lifecycle of recording a performance.

R1 — REC_IDLE

Purpose

Recorder is inactive and waiting for user input.

Entry

- System boots.
- Recording finishes.
- Recording cancelled.

Responsibilities

- Display recorder ready status.
- Wait for Record button.
- Keep recorder buffers empty.

Valid Transition

→ REC_ARMED

Exit

User requests recording.

R2 — REC_ARMED

Purpose

Prepare the recorder before the first note is captured.

Entry

Record button pressed.

Responsibilities

- Create recording session.
- Allocate RAM buffers.
- Generate file name.
- Prepare SD write structures.

Valid Transition

→ REC_WAIT_FIRST_NOTE

→ REC_IDLE

Exit

Recorder ready or user cancels.

R3 — REC_WAIT_FIRST_NOTE

Purpose

Recorder waits for the musician's first musical action.

Entry

Recorder successfully armed.

Responsibilities

- Timer prepared but not started.

- Ignore silence.
- Wait for first key event.

Valid Transition

→ REC_RECORDING

Exit

First musical event detected.

R4 — REC_RECORDING

Purpose

Capture every performance event.

Entry

First key pressed.

Responsibilities

- Start timer.
- Record note on/off events.
- Record transpose changes.
- Record accompaniment events.
- Update LCD timer.

Valid Transition

→ REC_PAUSED

→ REC_STOPPING

Exit

Pause or Stop requested.

R5 — REC_PAUSED

Purpose

Temporarily suspend recording.

Entry

Pause button pressed.

Responsibilities

- Preserve current session.
- Freeze timer.
- Keep recorded data intact.

Valid Transition

→ REC_RECORDING

→ REC_STOPPING

Exit

Resume or Stop.

R6 — REC_STOPPING

Purpose

Finish recording safely.

Entry

Stop requested.

Responsibilities

- Close event stream.
- Finalize metadata.
- Stop timer.

Valid Transition

→ REC_WRITING

Exit

Recording finalized.

R7 — REC_WRITING

Purpose

Write recording to SD card.

Entry

Recording finalized.

Responsibilities

- Write event data.
- Save metadata.
- Close file.

Valid Transition

→ REC_VERIFY

Exit

File written.

R8 — REC_VERIFY

Purpose

Ensure recording was successfully stored.

Entry

File writing completed.

Responsibilities

- Verify file exists.
- Verify file size.
- Report success or failure.

Valid Transition

→ REC_COMPLETE

→ REC_IDLE

Exit

Verification completed.

R9 — REC_COMPLETE

Purpose

Notify user recording finished successfully.

Entry

Verification successful.

Responsibilities

- Display recording complete.
- Show filename.
- Return controls to user.

Valid Transition

→ REC_IDLE

Exit

User acknowledges completion.

Runtime State Machine 4 — Playback

Playback is intentionally separate from recording.

P1 — PLAY_IDLE

Purpose

Playback subsystem inactive.

Entry

System startup or playback completion.

Responsibilities

- Wait for playback request.
- Keep playback buffers empty.

Valid Transition

→ PLAY_SELECT

Exit

Playback requested.

P2 — PLAY_SELECT

Purpose

Choose recording to play.

Entry

Playback button pressed.

Responsibilities

- Browse recordings.
- Display filenames.
- Allow selection.

Valid Transition

→ PLAY_LOADING

→ PLAY_IDLE

Exit

File selected or cancelled.

P3 — PLAY_LOADING

Purpose

Load playback file.

Entry

Recording selected.

Responsibilities

- Open file.
- Load metadata.
- Prepare playback buffers.

Valid Transition

→ PLAY_READY

Exit

Recording loaded.

P4 — PLAY_READY

Purpose

Playback prepared but not started.

Entry

Recording loaded.

Responsibilities

- Display duration.
- Wait for Play command.
- Initialize playback timer.

Valid Transition

→ PLAY_RUNNING

→ PLAY_IDLE

Exit

Playback starts or cancelled.

P5 — PLAY_RUNNING

Purpose

Replay the recorded performance.

Entry

Play pressed.

Responsibilities

- Replay note events.
- Drive keyboard LEDs.
- Update playback timer.
- Maintain synchronization.

Valid Transition

→ PLAY_PAUSED

→ PLAY_STOPPING

→ PLAY_FINISHED

Exit

Pause, Stop, or end of file.

P6 — PLAY_PAUSED

Purpose

Temporarily halt playback.

Entry

Pause pressed.

Responsibilities

- Freeze playback timer.
- Preserve playback position.

Valid Transition

→ PLAY_RUNNING

→ PLAY_STOPPING

Exit

Resume or Stop.

P7 — PLAY_STOPPING

Purpose

Terminate playback safely.

Entry

Stop requested.

Responsibilities

- Stop scheduled events.
- Turn off active notes.
- Clear LED indicators.

Valid Transition

→ PLAY_IDLE

Exit

Playback fully stopped.

P8 — PLAY_FINISHED

Purpose

Natural completion of playback.

Entry

End of recording reached.

Responsibilities

- Stop timer.
- Turn off LEDs.
- Display playback complete.

Valid Transition

→ PLAY_IDLE

Exit

User returns to normal operation.

Summary

This document defines **17 actual firmware runtime states**.

Recorder

- REC_IDLE
- REC_ARMED
- REC_WAIT_FIRST_NOTE
- REC_RECORDING
- REC_PAUSED
- REC_STOPPING
- REC_WRITING
- REC_VERIFY
- REC_COMPLETE

Playback

- PLAY_IDLE
- PLAY_SELECT
- PLAY_LOADING
- PLAY_READY
- PLAY_RUNNING
- PLAY_PAUSED
- PLAY_STOPPING
- PLAY_FINISHED

These states will become dedicated recorder and playback state machines inside the ESP32 firmware and operate independently of the audio synthesis engine.